JALA Academy
Assignment 15

Dictionaries in Python

Python solution reference • Code examples are organized by the assignment questions.

How to use this document

Save the code examples into a .py file and run with Python 3. Where a program asks for input, enter values when prompted. For file-handling exercises, create the named text file in the same folder first.

1–1.9. Create, update, access, iterate, nested data and delete

```
students = {
    101: "Anu",
    102: "Ravi",
    103: "Meena",
    104: "Kiran",
    105: "Sara",
}

# Add new entries and update a value.
students[106] = "Arun"
students[102] = "Ravi Kumar"

print("Student 103:", students.get(103))
print("All names:", list(students.values()))

for student_id, name in students.items():
    print(student_id, "->", name)

print("Keys:", list(students.keys()))
print("Values:", list(students.values()))

details = {
    101: {"name": "Anu", "age": 22},
    102: {"name": "Ravi Kumar", "age": 21},
}
print("Nested name:", details[101]["name"])
print("Nested age:", details[101]["age"])

del students[104]
students.popitem()  # Removes the last inserted entry.
students.pop(105, None)
```

2–5. Key existence, count, merge and dictionary comprehension

```
students_a = {101: "Anu", 102: "Ravi"}
students_b = {103: "Meena", 104: "Kiran"}

print("ID 101 exists:", 101 in students_a)
print("Number of students:", len(students_a))

combined = students_a | students_b  # Python 3.9+
print("Merged:", combined)

squares = {number: number ** 2 for number in range(1, 6)}
print("Squares:", squares)
```